

Homework #2

Noah Arias

Problem 1

Part (a)

Let $S = AA^\top$. Thus:

$$S = AA^\top = \begin{bmatrix} 4 & 8 & 8 \\ 3 & 6 & -9 \end{bmatrix} \cdot \begin{bmatrix} 4 & 3 \\ 8 & 6 \\ 8 & -9 \end{bmatrix} = \begin{bmatrix} 144 & -12 \\ -12 & 126 \end{bmatrix}$$

(We briefly pause the calculation to note that S this matrix is symmetric and non-singular.)

To find the eigenvalues of S , we create and solve the characteristic equation:

$$\det(S - \lambda I_2) = 0$$

where I_2 is the 2×2 identity matrix. Performing the calculation, we get:

$$\begin{aligned} \det(S - \lambda I_2) &= \det\left(\begin{bmatrix} 144 & -12 \\ -12 & 126 \end{bmatrix} - \lambda \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) \\ &= \det\left(\begin{bmatrix} 144 - \lambda & -12 \\ -12 & 126 - \lambda \end{bmatrix}\right) \\ &= (144 - \lambda)(126 - \lambda) - 144 \\ &= 18,144 - 144\lambda - 126\lambda + \lambda^2 - 144 \\ &= \lambda^2 - 270\lambda + 18,000 \\ &= 0 \end{aligned}$$

Solving this quadratic equation with the quadratic formula yields the two eigenvalues, λ_1 and λ_2 being:

$$\lambda_1 = 150, \lambda_2 = 120$$

$\lambda_1 = 150$	$\lambda_2 = 120$
$(S - \lambda_1 I_2) \cdot \vec{v} = 0$	$(S - \lambda_2 I_2) \cdot \vec{v} = 0$
$\begin{bmatrix} 144 - 150 & -12 \\ -12 & 126 - 150 \end{bmatrix} = \begin{bmatrix} -6 & -12 \\ -12 & -24 \end{bmatrix}$	$\begin{bmatrix} 144 - 120 & -12 \\ -12 & 126 - 120 \end{bmatrix} = \begin{bmatrix} 24 & -12 \\ -12 & 6 \end{bmatrix}$
$\Rightarrow \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$	$\Rightarrow \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix}$

$$\lambda_1 = 150$$

$$\lambda_2 = 120$$

Thus $x_1 + 2x_2 = 0 \implies x_1 = -2x_2$. The eigenvector for λ_1 is $(-2, 1)$. Thus $2x_1 - x_2 = 0 \implies x_2 = 2x_1$. The eigenvector for λ_1 is $(1, 2)$.

Part (b) WIP!

Let $S = AA^T$. Thus:

$$S = AA^T = \begin{bmatrix} 4 & 8 & 8 \\ 3 & 6 & -9 \end{bmatrix} \cdot \begin{bmatrix} 4 & 3 \\ 8 & 6 \\ 8 & -9 \end{bmatrix} = \begin{bmatrix} 144 & -12 \\ -12 & 126 \end{bmatrix}$$

(We briefly pause the calculation to note that S this matrix is symmetric and non-singular.)

To find the eigenvalues of S , we create and solve the characteristic equation:

$$\det(S - \lambda I_2) = 0$$

where I_2 is the 2×2 identity matrix. Performing the calculation, we get:

$$\begin{aligned} \det(S - \lambda I_2) &= \det\left(\begin{bmatrix} 144 & -12 \\ -12 & 126 \end{bmatrix} - \lambda \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) \\ &= \det\left(\begin{bmatrix} 144 - \lambda & -12 \\ -12 & 126 - \lambda \end{bmatrix}\right) \\ &= (144 - \lambda)(126 - \lambda) - 144 \\ &= 18,144 - 144\lambda - 126\lambda + \lambda^2 - 144 \\ &= \lambda^2 - 270\lambda + 18,000 \\ &= 0 \end{aligned}$$

Solving this quadratic equation with the quadratic formula yields the two eigenvalues, λ_1 and λ_2 being:

$$\lambda_1 = 150, \lambda_2 = 120$$

$$\lambda_1 = 150$$

$$\lambda_2 = 120$$

$$\begin{array}{cc} (S - \lambda_1 I_2) \cdot \vec{v} = 0 & (S - \lambda_2 I_2) \cdot \vec{v} = 0 \\ \begin{bmatrix} 144 - 150 & -12 \\ -12 & 126 - 150 \end{bmatrix} = \begin{bmatrix} -6 & -12 \\ -12 & -24 \end{bmatrix} & \begin{bmatrix} 144 - 120 & -12 \\ -12 & 126 - 120 \end{bmatrix} = \begin{bmatrix} 24 & -12 \\ -12 & 6 \end{bmatrix} \\ \implies \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} & \implies \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix} \end{array}$$

Thus $x_1 + 2x_2 = 0 \implies x_1 = -2x_2$. The eigenvector for λ_1 is $(-2, 1)$. Thus $2x_1 - x_2 = 0 \implies x_2 = 2x_1$. The eigenvector for λ_1 is $(1, 2)$.